

Lecture 5: Predictor-Corrector Interior Point Methods

Algorithms & Geometry of Linear Programming

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Where we stand

Lecture 4: what the geometry of the central path *allows*.

- every path-following method needs $\geq \max_i \text{SLC}(x_i^m)$ iterations — and this is $2^{\Omega(n)}$ on tropical instances [ABGJ '18]
- yet $\text{SLC}_{1/(n\kappa_{\mathbf{A}})^2}(x_i^m) \leq \min\{m, n - m\} + 2$ when the circuit imbalance $\kappa_{\mathbf{A}}$ is bounded [DKNOV '24]

What remains is **the algorithm**. Today:

1. the classical **predictor-corrector** IPM and its $O(\sqrt{n} \log(1/\varepsilon))$ bound [Mizuno–Todd–Ye '93]
2. its structural weakness — **no finite termination**, even on a straight path — and the **trust-region step** that repairs it, matching the straight line complexity [ADLNV '22]

Today

The Primal-Dual Central Path

The Predictor-Corrector Method

Reaching the End of the Path: the Trust-Region Step

Outline

The Primal-Dual Central Path

The Predictor-Corrector Method

Reaching the End of the Path: the Trust-Region Step

The primal-dual pair

Standard form primal and dual

$$\min c^T x \text{ s.t. } Ax = b, x \geq 0$$

$$\max b^T y \text{ s.t. } A^T y + s = c, s \geq 0$$

$A \in \mathbb{R}^{m \times n}$ of full row rank; both programs *strictly* feasible. Write $z = (x, s, y)$.

- **gap formula:** for feasible z : $c^T x - b^T y = s^T x \geq 0$
- **complementary slackness:** $\text{gap } 0 \iff x_i s_i = 0 \forall i$ — then both sides optimal
- **notation:** products of vectors are coordinatewise: $xs := (x_1 s_1, \dots, x_n s_n)$,
likewise $x/s, \sqrt{x}, x^{-1}$

Interior point methods relax $x_i s_i = 0$ to $x_i s_i = \mu > 0$ and drive $\mu \rightarrow 0$.

Two barrier problems

The primal log-barrier problem of Lecture 4, and its dual counterpart:

$$x(\mu) := \operatorname{argmin} \left\{ c^\top x - \mu \sum_{i=1}^n \ln x_i : \mathbf{A}x = b, x > \mathbf{0} \right\}$$

$$(s(\mu), y(\mu)) := \operatorname{argmax} \left\{ b^\top y + \mu \sum_{i=1}^n \ln s_i : \mathbf{A}^\top y + s = c, s > \mathbf{0} \right\}$$

- both optima exist and are unique for every $\mu > 0$: strict convexity/concavity + strict feasibility, as in Lecture 4
- μ trades objective against distance to the boundary: $\mu \rightarrow \infty$ gives the *analytic center*, $\mu \rightarrow 0$ the optimum

The central path equations

Lemma (Central path equations)

For every $\mu > 0$, the point $z(\mu) := (x(\mu), s(\mu), y(\mu))$ is uniquely characterized by

$$\mathbf{A}x = b, \quad x > \mathbf{0}, \quad \mathbf{A}^\top y + s = c, \quad s > \mathbf{0}, \quad xs = \mu \mathbf{1}.$$

relaxed complementary slackness: every product $x_i s_i$ equals μ , not 0

Proof, existence. $x := x(\mu)$ is an *interior* minimizer of the convex $f(w) := c^\top w - \mu \sum_i \ln w_i$ over $\{w : \mathbf{A}w = b\}$:

- first-order optimality: $\nabla f(x) = c - \mu x^{-1} \perp \ker(\mathbf{A}) = \text{im}(\mathbf{A}^\top)^\perp$, so $c - \mu x^{-1} = \mathbf{A}^\top y$ for some $y \in \mathbb{R}^m$
- set $s := \mu x^{-1} > \mathbf{0}$: then $\mathbf{A}^\top y + s = c$ and $xs = \mu \mathbf{1}$

□

One system, two barrier problems

- **uniqueness:** a solution (x', s', y') has $\nabla f(x') = c - \mu(x')^{-1} = \mathbf{A}^\top y' \perp \ker(\mathbf{A})$, so x' minimizes f : $x' = x(\mu)$; then $s' = \mu(x')^{-1}$ and y' are determined ($\mathbf{A}$ full row rank)
- **the dual barrier gives the same system:** substitute $s = c - \mathbf{A}^\top y$ and set the gradient in y to zero:

$$b - \mathbf{A}(\mu s^{-1}) = \mathbf{0},$$

the *same* three equations, now with $x := \mu s^{-1}$

- hence the dual maximizer is the $(s(\mu), y(\mu))$ of the same triple $z(\mu)$

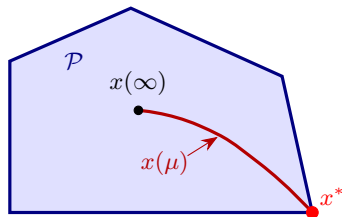
The primal-dual central path

- $\mu \mapsto z(\mu)$: the **primal-dual central path**; its primal part is the path of Lecture 4
- summing $x_i s_i = \mu$: gap on the path

$$c^T x(\mu) - b^T y(\mu) = n\mu$$

so $\mu \rightarrow 0$ solves *both* programs

- $z^* := \lim_{\mu \rightarrow 0} z(\mu)$ exists, is **strictly complementary** [Goldman–Tucker '56]: exactly one of $x_i^* > 0$, $s_i^* > 0$ per coordinate
- *optimality partition* $B \cup N = [n]$:
 $B = \text{supp}(x^*)$, $N = \text{supp}(s^*)$



the primal picture; the dual path $s(\mu)$ is coupled by $xs = \mu \mathbf{1}$

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Reaching the End of the Path: the Trust-Region Step

Staying near the path

Iterates: strictly feasible points, close to the path *multiplicatively*.

Definition (Normalized gap, neighborhood)

For strictly feasible $z = (x, s, y)$:

$$\mu(z) := \frac{x^T s}{n}, \quad N_2(\beta) := \left\{ z : \left\| \frac{xs}{\mu(z)} - \mathbf{1} \right\|_2 \leq \beta \right\}.$$

- on the path $xs = \mu \mathbf{1}$; $\overline{N_2}(\beta) := \text{closure (contains } z^*)$
- since $\ell_\infty \leq \ell_2$: in $N_2(\beta)$,

$$(1 - \beta) \mu(z) \leq x_i s_i \leq (1 + \beta) \mu(z) \quad \forall i \in [n]$$

Throughout: $\beta \in (0, 1/2]$; think $\beta = 1/4$.

One linearization, two targets

Move $z \rightarrow z + \Delta z$ keeping the linear constraints. The new products:

$$(x + \Delta x)(s + \Delta s) = xs + \underbrace{s \Delta x + x \Delta s}_{\text{linear in } \Delta z} + \Delta x \Delta s.$$

Dropping the quadratic term: the **Newton step** towards target products $xs + t$ solves

$$s \Delta x + x \Delta s = t, \quad \mathbf{A} \Delta x = \mathbf{0}, \quad \mathbf{A}^\top \Delta y + \Delta s = \mathbf{0}.$$

- **corrector:** $t = \mu(z)\mathbf{1} - xs$ — aim at the path point with the *current* gap
- **predictor:** $t = -xs$ — aim straight at the optimum (a.k.a. *affine scaling*)
- in both cases the **entire error** of the step is the neglected term $\Delta x \Delta s$

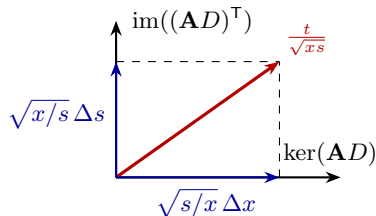
The steps as projections

- the subspace constraints put $\Delta x \in \ker(\mathbf{A})$, $\Delta s \in \text{im}(\mathbf{A}^\top) = \ker(\mathbf{A})^\perp$: $\Delta x^\top \Delta s = 0$, always
- divide $s\Delta x + x\Delta s = t$ by $\sqrt{xs}$; with $D := \text{diag}(\sqrt{x/s})$:

$$\underbrace{\sqrt{s/x} \Delta x}_{\in \ker(\mathbf{A}D)} + \underbrace{\sqrt{x/s} \Delta s}_{\in \text{im}((\mathbf{A}D)^\top)} = t/\sqrt{xs}$$

- an **orthogonal decomposition** of $t/\sqrt{xs}$: the step exists, is unique, costs one linear system solve

Pythagoras: $\left\| \sqrt{s/x} \Delta x \right\|_2^2 + \left\| \sqrt{x/s} \Delta s \right\|_2^2 = \|t/\sqrt{xs}\|_2^2$



Two elementary facts do all the work

Fact 1. For any $u, v \in \mathbb{R}^n$ (coordinatewise product):

$$\|uv\|_2 \leq \|u\|_2 \|v\|_2 \leq \frac{1}{2} (\|u\|_2^2 + \|v\|_2^2)$$

Fact 2 (exact gap evolution). Since $\Delta x^\top \Delta s = 0$:

$$n \mu(z + \alpha \Delta z) = (x + \alpha \Delta x)^\top (s + \alpha \Delta s) = x^\top s + \alpha \mathbf{1}^\top t.$$

- **corrector** ($t = \mu(z)\mathbf{1} - xs$): $\mathbf{1}^\top t = 0$ — gap **unchanged**
- **predictor** ($t = -xs$): $\mathbf{1}^\top t = -x^\top s$ — gap **scaled by exactly $(1 - \alpha)$**

The predictor-corrector method

PREDICTOR-CORRECTOR IPM

[Mizuno–Todd–Ye '93]

Input: $z \in N_2(\beta)$, $\beta \in (0, 1/2]$, target $\varepsilon \geq 0$.

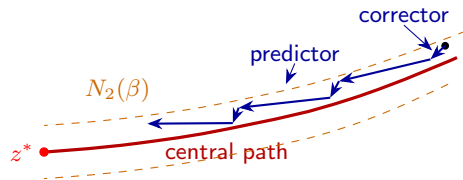
$\text{DIR}(z, t)$: the unique Δz with

$$s \Delta x + x \Delta s = t, \quad \mathbf{A} \Delta x = \mathbf{0}, \quad \mathbf{A}^\top \Delta y + \Delta s = \mathbf{0}.$$

while $x^\top s > \varepsilon$:

- *corrector*: $\Delta z^c \leftarrow \text{DIR}(z, \mu(z)\mathbf{1} - xs)$;
 $z \leftarrow z + \Delta z^c$
- *predictor*: $\Delta z^p \leftarrow \text{DIR}(z, -xs)$;
largest α with $[z, z + \alpha \Delta z^p] \subseteq \overline{N_2(\beta)}$;
 $z \leftarrow z + \alpha \Delta z^p$

return z



The corrector re-centers quadratically

Lemma (Corrector)

Let $z \in N_2(\beta)$, $\beta \in (0, 1/2]$. Then the gap is preserved, $\mu(z + \Delta z^c) = \mu(z)$, and

$$z + \Delta z^c \in N_2\left(\frac{\beta^2}{2(1-\beta)}\right) \subseteq N_2(\beta/2).$$

Proof idea. The target equation makes the new products $\mu \mathbf{1} + \Delta x^c \Delta s^c$: the deviation is *exactly* the quadratic term. Fact 1 + Pythagoras bound it by the squared step length:

$$\left\| \frac{(x + \Delta x^c)(s + \Delta s^c)}{\mu} - \mathbf{1} \right\|_2 = \left\| \frac{\Delta x^c \Delta s^c}{\mu} \right\|_2 \leq \frac{1}{2} \left\| \frac{\mu \mathbf{1} - xs}{\sqrt{\mu xs}} \right\|_2^2 \leq \frac{\beta^2}{2(1-\beta)}$$

using $x_i s_i \geq (1 - \beta) \mu$, then $\left\| \frac{\mu \mathbf{1} - xs}{\mu} \right\|_2 \leq \beta$. □

The predictor advances by $\Omega(1/\sqrt{n})$

Lemma (Predictor)

Let $z \in N_2(\beta/2)$, $\beta \in (0, 1/2]$. Then:

1. $\mu(z + \alpha \Delta z^p) = (1 - \alpha) \mu(z)$ for all α ;
2. $\|\Delta x^p \Delta s^p / \mu(z)\|_2 \leq n/2$;
3. any $\alpha \in [0, 1)$ with $\frac{\alpha^2}{1-\alpha} \|\Delta x^p \Delta s^p / \mu(z)\|_2 \leq \frac{\beta}{2}$ keeps $z + \alpha \Delta z^p \in N_2(\beta)$; in particular $\alpha \geq \frac{1}{2} \sqrt{\beta/n}$ is accepted.

Proof idea.

- (2): the corrector computation with $t = -xs$:

$$\left\| \frac{\Delta x^p \Delta s^p}{\mu} \right\|_2 \leq \frac{1}{2} \left\| \frac{xs}{\sqrt{\mu} xs} \right\|_2^2 = \frac{1}{2} \sum_i \frac{x_i s_i}{\mu} = \frac{n}{2}$$

- (3): new products $= (1 - \alpha) xs + \alpha^2 \Delta x^p \Delta s^p$; normalizing by the new gap $(1 - \alpha)\mu$, the deviation from **1** is $\leq \frac{\beta}{2} + \frac{\alpha^2}{1-\alpha} \left\| \frac{\Delta x^p \Delta s^p}{\mu} \right\|_2$; plug in $\alpha = \frac{1}{2} \sqrt{\beta/n}$ $\square$

The iteration bound

Theorem (Mizuno–Todd–Ye '93)

From $z_0 \in N_2(\beta)$, $\beta \in (0, 1/2]$, the method stays in $\overline{N_2}(\beta)$ and reaches gap ε within

$$2\sqrt{n/\beta} \ln \left(\frac{x_0^\top s_0}{\varepsilon} \right) + 1 \quad \text{iterations.}$$

Proof. Induction: corrector $N_2(\beta) \rightarrow N_2(\beta/2)$ at equal gap; predictor multiplies the gap by $(1 - \alpha) \leq e^{-\frac{1}{2}\sqrt{\beta/n}}$, returning to $\overline{N_2}(\beta)$. □

- each iteration: $O(1)$ linear system solves
- β constant: the gap **halves every $O(\sqrt{n})$ iterations** — Renegar's bound, now self-contained
- over $\mathbb{Q}$: run to $\varepsilon = 2^{-\Theta(L)}$ and round: exact optimum in $O(\sqrt{n} L)$ iterations

Initialization: the homogeneous self-dual embedding [Ye–Todd–Mizuno '94]

Let $\bar{b} := b - \mathbf{A}\mathbf{1}$, $\bar{c} := c - \mathbf{1}$, $\bar{g} := c^\top \mathbf{1} + 1$:

$$\begin{aligned} \min (n+1)\theta \text{ s.t. } & \mathbf{A}x - b\tau + \bar{b}\theta = \mathbf{0}, \quad -\mathbf{A}^\top y + c\tau - \bar{c}\theta = s \geq \mathbf{0}, \quad b^\top y - c^\top x + \bar{g}\theta = \kappa \geq 0, \\ & -\bar{b}^\top y + \bar{c}^\top x - \bar{g}\tau = -(n+1), \quad x \geq \mathbf{0}, \tau \geq 0. \end{aligned}$$

- $(y, x, \tau, \theta) = (\mathbf{0}, \mathbf{1}, 1, 1)$ is feasible with slacks $(s, \kappa) = (\mathbf{1}, 1)$: **perfectly centered** — start there
- skew-symmetric matrix $\Rightarrow$ **self-dual**; every feasible point: $x^\top s + \tau\kappa = (n+1)\theta$ — the objective *is* the gap
- a strictly complementary optimum ($\theta = 0$) *decides* the LP: $\tau > 0$ gives an optimal pair $(x/\tau, s/\tau, y/\tau)$; $\kappa > 0$ gives a Farkas certificate of infeasibility

Recap of the first hour

- **primal-dual central path:** $\mathbf{A}x = b$, $\mathbf{A}^\top y + s = c$, $x, s > \mathbf{0}$, $xs = \mu \mathbf{1}$; $\text{gap} = n\mu$
- **Newton on the products:** corrector re-centers quadratically; predictor shrinks the gap by $(1 - \alpha)$, $\alpha \geq \frac{1}{2}\sqrt{\beta/n}$
- **[MTY '93]:** gap ε in $O(\sqrt{n} \log(1/\varepsilon))$ iterations
- **self-dual embedding:** centered start, no feasibility assumptions

But: the while-loop tests $x^\top s > \varepsilon$ — and for $\varepsilon = 0$ it **never exits**.

Second hour: why the predictor cannot finish — even when the remaining path is straight — and the step that can.

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The finite-termination obstruction

- the gap decreases geometrically — but remains positive **forever**
- exact optima over $\mathbb{Q}$: stop at $\varepsilon = 2^{-\Theta(L)}$ and round — but the threshold depends on the **bit size** L , which a strongly polynomial algorithm cannot use
- the goal set by Lecture 4 — iteration bounds matching SLC, strongly polynomial for bounded κ_A — requires the method to **reach the end of the path**
- yet the end of the path is the **easiest possible geometry**: asymptotically a straight line

Plan: examine the last stretch of the path, understand why the predictor cannot walk it, then design the step that can.

The last stretch, up close

Normalization (exercise): rescaling coordinates maps central path to central path; wlog the current point is $z(1)$ with $(x(1), s(1)) = (\mathbf{1}, \mathbf{1})$, $\mu = 1$.

Assume the path has essentially converged in shape: with $B \cup N$ the optimality partition $(x_N^* = \mathbf{0}, s_B^* = \mathbf{0})$,

$$\|(x_B^* - \mathbf{1}_B, s_N^* - \mathbf{1}_N)\|_2 \leq \beta.$$

The one step to the end, $\Delta z := z^* - z(1)$, has the **polarization pattern**:

$$\Delta x = \left(\underbrace{\Delta x_B}_{\|\cdot\|_2 \leq \beta}, \underbrace{\Delta x_N}_{=-\mathbf{1}_N} \right) \approx (\mathbf{0}_B, -\mathbf{1}_N), \quad \Delta s = \left(\underbrace{\Delta s_B}_{=-\mathbf{1}_B}, \underbrace{\Delta s_N}_{\|\cdot\|_2 \leq \beta} \right) \approx (-\mathbf{1}_B, \mathbf{0}_N).$$

The final segment lies inside the tube

Proposition (The final segment is central)

Under the closeness assumption, the segment $z(1) + \alpha\Delta z$, $\alpha \in [0, 1]$, satisfies

$$(\mathbf{1} + \alpha\Delta x)(\mathbf{1} + \alpha\Delta s) = (1 - \alpha)(\mathbf{1} + \alpha e), \quad e := (\Delta x_B, \Delta s_N), \quad \|e\|_2 \leq \beta,$$

so after normalizing the gap by $(1 - \alpha)$, the centrality distance along the segment is at most $\alpha\beta \leq \beta$.

Proof idea. Blockwise: on N , $\Delta x_N = -\mathbf{1}_N$ gives the factor $(1 - \alpha)$ exactly, leaving $(1 + \alpha\Delta s_N)$; on B , the mirror image. $\square$

The ideal step travels the *entire* remaining path inside the tube, reaching the exact optimum at $\alpha = 1$.

Why the predictor cannot take this step

At $(x, s) = (1, 1)$ the affine scaling equation $s\Delta x + x\Delta s = -xs$ reads

$$\Delta x_i + \Delta s_i = -1 \quad \text{coordinatewise:}$$

each coordinate *splits* the decrease -1 between primal and dual side.

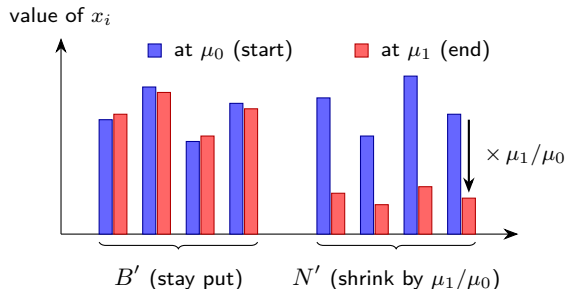
- the polarized step demands, on $i \in B$: $(\Delta x_i, \Delta s_i) \approx (0, -1)$ — product error $|\Delta x_i \Delta s_i| \approx |\Delta x_i|$, tiny
- consider the pattern $(\Delta x_i, \Delta s_i) = (-1 + \nu, \beta)$, $\nu \ll 1$: product $\nu(1 + \beta)$ — consistent with $N_2(\beta)$ at gap $\nu \dots$
- ... but its quadratic term is $|\Delta x_i \Delta s_i| \approx \beta \gg \beta\nu$, and the predictor's error **equals** its quadratic term: out of reach

$\Rightarrow$ bounded gap decrease per step; **no finite termination** (exercise).

Polarized segments: where long steps are possible

The piece of the path between $\mu_1 < \mu_0$ is **polarized** if for some partition $B' \cup N' = [n]$, on the whole piece:

$$x_{B'}(\mu) \approx x_{B'}(\mu_0), \quad x_{N'}(\mu) \approx \frac{\mu}{\mu_0} x_{N'}(\mu_0), \quad \text{roles swapped for } s.$$



each coordinate stays put or scales with the gap

The trust-region step

The step must be *told* the partition, and asked for **asymmetric** motion: shrink $x_{N'}$ as much as possible, disturb $x_{B'}$ as little as possible — and dually.

Trust-region problems

[Lan–Monteiro–Tsuchiya '09]

$$\Delta x^{\text{tr}} := \operatorname{argmin} \left\{ \left\| \frac{x_{N'} + \Delta x_{N'}}{x_{N'}} \right\|_2 : \mathbf{A} \Delta x = \mathbf{0}, \left\| \frac{\Delta x_{B'}}{x_{B'}} \right\|_2 \leq \frac{1}{10} \right\}$$
$$\Delta s^{\text{tr}} := \operatorname{argmin} \left\{ \left\| \frac{s_{B'} + \Delta s_{B'}}{s_{B'}} \right\|_2 : \Delta s \in \operatorname{im}(\mathbf{A}^\top), \left\| \frac{\Delta s_{N'}}{s_{N'}} \right\|_2 \leq \frac{1}{10} \right\}$$

- minimize the relative residual of the coordinates that should *vanish*, inside a **trust region** protecting the coordinates that should *stay put*
- convex quadratic over ball $\cap$ subspace: a few linear system solves

The trust-region guarantee

Lemma (Lan–Monteiro–Tsuchiya '09; ADLNV '22)

Let $z \in N_2(1/10)$ and let $r := \max \left\{ \left\| \frac{x_{N'} + \Delta x_{N'}^{\text{tr}}}{x_{N'}} \right\|_2, \left\| \frac{s_{B'} + \Delta s_{B'}^{\text{tr}}}{s_{B'}} \right\|_2 \right\}$. If $r \leq 1/10$, then

$$z + (1 - 10r) \Delta z^{\text{tr}} \in N_2(1/2), \quad \mu(z + (1 - 10r) \Delta z^{\text{tr}}) = O(r) \mu(z).$$

- in one step, the gap shrinks by the largest factor any centrally-confined motion with this partition permits
- on a polarized segment down to μ_1 : residual $r = O(\mu_1/\mu_0)$ — a **single** trust-region step (plus correctors) traverses the segment
- at the end of the path the residual reaches $r = 0$: the step lands **exactly on z^*** — finite termination, no rounding

Guessing the partition

One ingredient missing: who supplies (B', N') ?

The predictor direction — the step that cannot walk the segment — **reveals** it: affine scaling splits -1 between $\Delta x_i^p/x_i$ and $\Delta s_i^p/s_i$ on each coordinate, and on a polarized segment the split is lopsided in the direction of the polarization.

Read off which side takes the larger share of the decrease:

$$N' := \left\{ i : \frac{\Delta x_i^p}{x_i} < \frac{\Delta s_i^p}{s_i} \right\}, \quad B' := [n] \setminus N'.$$

The trust-region IPM

TRUST-REGION IPM

[Lan–Monteiro–Tsuchiya '09]

Input: $z \in N_2(\beta)$, β a small constant.

while $x^T s > 0$:

- corrector step at z
- predictor step: compute Δz^p and the maximal step α
- **if** $\alpha \geq 3/4$: guess (B', N') from Δz^p ; solve the trust-region problems; take the trust-region step if it improves on $z + \alpha \Delta z^p$, else the predictor step
- **else:** take the predictor step $z \leftarrow z + \alpha \Delta z^p$

return z

- a long predictor step ($\alpha \geq 3/4$) signals a polarized stretch ahead
- the trust-region step is taken only when it *beats* the predictor: never slower than [MTY '93]

Iterations match the straight line complexity

Theorem (ADLNV '22)

Started at a well-centered z_0 with gap g_0 , the TRUST-REGION IPM computes an exact optimal pair $z^ \in \overline{N}_2(\beta)$ within*

$$O\left(\min_{\gamma \in (0,1]} \sqrt{n} \log\left(\frac{n}{\gamma}\right) \sum_{i=1}^n \text{SLC}_{\gamma}(x_i^{\text{m}}, g_0)\right) \quad \text{iterations.}$$

Proof outline

- decompose the (max) central path into $\sum_i \text{SLC}_{\gamma}(x_i^{\text{m}})$ approximately polarized segments (Lecture 4 + exercise)
- each segment is traversed in $O(\sqrt{n} \log(n/\gamma))$ iterations (not proved in course)

Literature on the Trust-Region IPM

- the trust-region IPM and the trust-region problems were developed by **Lan, Monteiro and Tsuchiya '09**
- the SLC iteration bound is *implicit* in **ADLNV '22**: the algorithm analyzed there follows the same scheme, with a **very approximate** version of the trust-region step — enough for the SLC bound
- **D–Ma–Natura–Végh '26**: an efficient — **strongly polynomial** — implementation of the trust-region step itself, and **optimality for ℓ_2 path following**: iteration count within a constant factor of *any* algorithm whose iterates stay in the same ℓ_2 neighborhood of the central path

Consequences

- **exact termination:** the Trust-Region IPM terminates at an optimal solution after a **finite** number of iterations, by following the central path until the end — it turns a classically approximate method into an exact one, just like the simplex method
- **matching the lower bound:** no path-following method can beat $\max_i \text{SLC}(x_i^m)$ iterations; the theorem achieves $\tilde{O}(\sqrt{n}) \sum_i \text{SLC}(x_i^m)$ — up to this polynomial factor, path-following IPMs are *exactly as fast as the geometry of the central path allows*; the tropical instances provide examples where this geometry itself can be exponential
- **LPs with strongly polynomial SLC:** with the circuit cover bound of Lecture 4,

$$O(n^{1.5} \min\{m, n - m\} \log(n \kappa_{\mathbf{A}})) \text{ iterations}$$

— strongly polynomial for bounded $\kappa_{\mathbf{A}}$, recovering Vavasis–Ye and Tardos-type results; for network matrices ($\kappa_{\mathbf{A}} = 1$): **min-cost flow admits strongly polynomial IPMs**; also any LP where $\mathbf{A}$ has at most two nonzero entries per column [DKNOV '24]

The week in review

- **Lecture 1** — the model: polyhedra, vertices, duality — each concept in a geometric and a linear-algebraic view
- **Lecture 2** — the simplex method; exponential pivot lower bounds [Klee–Minty '72]; the diameter of polyhedra [Kalai–Kleitman '92]
- **Lecture 3** — circuits, and smoothed analysis: why simplex is fast on perturbed instances
- **Lecture 4** — the central path and its geometry: straight line complexity, tropical lower bounds [ABGJ '18], circuit imbalance upper bounds [DKNOV '24]
- **Lecture 5** — the algorithms: predictor-corrector [MTY '93]; trust-region IPM with exact termination, matching SLC [ADLNV '22]

One storyline: the complexity of LP algorithms is governed by the geometry of paths — vertex walks for the simplex method, the central path for IPMs.

Open problems

- **the main open problem remains:** is there a strongly polynomial algorithm for all of linear programming?
- **straight line complexity:** which classes of linear programs have **strongly polynomial SLC**?
- **polyhedral diameter:** can one construct **super quadratic** lower bounds for connected layer families (CLFs)? — a purely combinatorial problem
- **reoptimization:** why is the simplex method so effective on the sequence of LPs arising in branch & bound? The simplex method on the dual receives a new variable each time a primal constraint is fixed, so it always reoptimizes from a feasible solution — in **far fewer** pivots than solving the original problem, so far without theoretical explanation

Exercises: the self-dual embedding, exact termination requires polarization, polarized segments and SLC.